

#17 YouTube Comment Insight Extractor

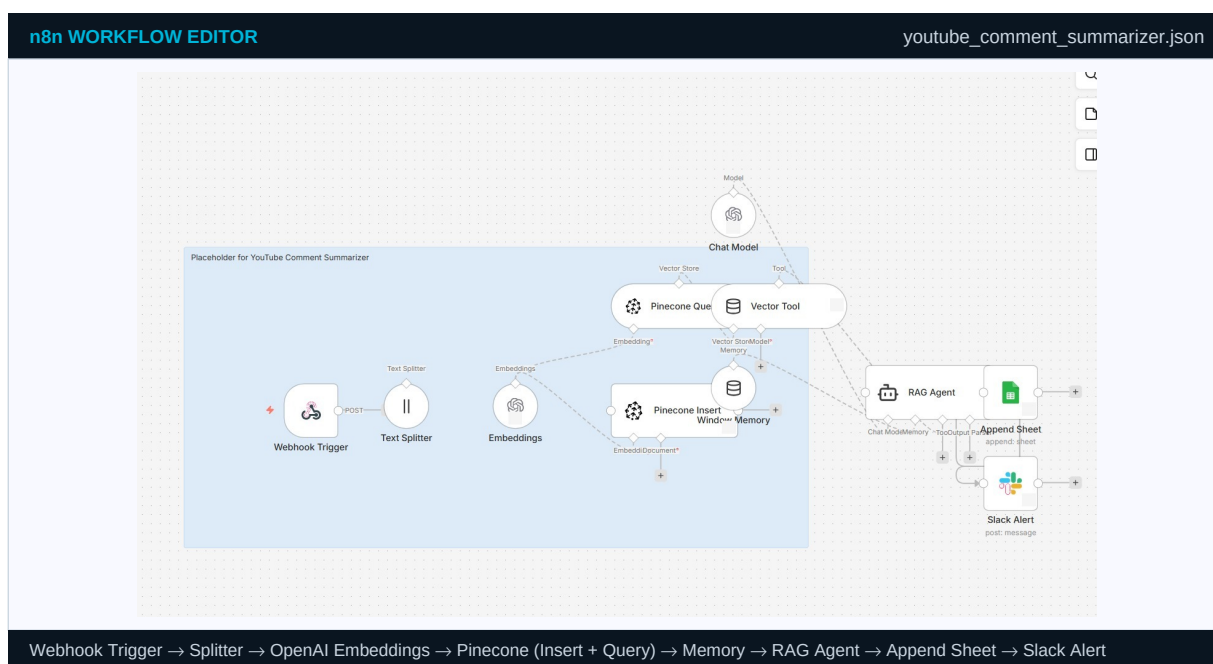
Video / YouTube
Intelligence

Mine 10,000 YouTube comments for audience pain points and feature requests

BUSINESS PROBLEM

A video with 10,000 comments contains invaluable audience feedback that nobody reads. This workflow ingests all comments, embeds them into Pinecone, clusters by theme, performs sentiment analysis at scale, and generates a structured insight report with representative quotes per theme.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — YouTube video URL for comment analysis

```
{
  "video_url": "https://youtube.com/watch?v=dQw4w9WgXcQ",
  "channel_id": "UC_channel_123",
  "analysis_type": "full",
  "max_comments": 10000,
  "date_range": {
    "from": "2024-01-01",
    "to": "2024-06-15"
  }
}
```

OUTPUT — Insight report — written to Sheets + Slack summary

```
{
  "json": {
    "video_url": "https://youtube.com/watch?v=dQw...",
    "comments_analysed": 8241,
    "overall_sentiment": 0.87,
    "nps_proxy": 72,
    "top_themes": [
      {"theme": "Need advanced examples", "mentions": 1840, "sentiment": 0.5},
      {"theme": "Audio quality issues", "mentions": 920, "sentiment": 0.18},
      {"theme": "Request Python integration", "mentions": 780, "sentiment": 0.72}
    ],
    "report_url": "https://docs.google.com/spreadsheets/d/1Bxi...",
    "processing_time_sec": 847
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — YouTube video URL	POST with {video_url, max_comments, date_range}. YouTube Data API v3 fetches all comments with pagination.
2	➤ Splitter — 50-comment chunks	Comments grouped into 50-comment chunks for efficient batch embedding. Average comment approximately 100 characters.
3	■ OpenAI Embeddings — comment vectors	Each comment embedded: similar complaints naturally cluster in vector space without any manual labelling.
4	■ Pinecone Insert — comment corpus	Up to 10,000 comments stored as vectors with metadata: {like_count, reply_count, published_at, author_id}.
5	■ Pinecone Query — theme discovery	Agent iteratively queries: 'comments about audio quality', 'feature requests', 'praise', 'complaints about length'.
6	■ Vector Tool — per-theme retrieval	Fetches top-50 comments per discovered theme cluster for GPT-4 synthesis into a readable summary.
7	■ Window Memory — analysis state	Tracks themes discovered so far. Prevents duplicate analysis of the same theme in iterative query loop.
8	■ RAG Agent — synthesise per theme	For each cluster: generates theme_name, sentiment_score (0-1), mention_count, top_3_representative_quotes.
9	■ Append Sheet — structured output	Table: {theme, mention_count, sentiment_score, top_quote, trend_direction}. Powers insight dashboard.
10	■ Slack — executive summary	Top 3 themes and overall sentiment score posted to product channel within 15 minutes of trigger.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
YouTube Data API v3	console.cloud.google.com → Enable YouTube Data API v3 → Create API key	Google API Key
OpenAI API Key	platform.openai.com	OpenAI API
Pinecone API Key	app.pinecone.io — create index: 1536 dims, cosine metric	Pinecone API
Slack Bot Token	api.slack.com/apps → OAuth & Permissions	Slack API

SAMPLE RUN — what this workflow processed and produced

Scenario	A developer education channel publishes weekly tutorials averaging 50,000 views and 400-800 comments per video. Product team wants to know what topics and formats the audience wants next.
Input received	Webhook triggered for a 45-minute n8n automation tutorial. YouTube API fetched 8,241 comments posted over 6 months. Most liked comment had 847 likes.

What the workflow found

Embedding 8,241 comments into Pinecone took 4 minutes 12 seconds. Vector clustering identified 23 meaningful themes. Top 3 by mention count: 'need advanced production examples' (1,840 mentions), 'audio quality concerns' (920), 'Python code node deep-dive request' (780).

Output produced

Full insight report in Google Sheets with 23 theme rows. Slack summary posted to #product-insights. Channel creator used results to plan next 3 videos: error handling deep-dive, improved recording setup, Python code node tutorial. All 3 videos exceeded average view count.

JSON: youtube_comment_summarizer.json**GitHub:** [trm-youtube-insight-agent](#)[developers.google.com/youtube/v3](#)